FOUNDATION TRAINING • THEORY AND MATHEMATICS

From Fourier transforms to a working OFDM link

Understand why the FFT and cyclic prefix make broadband wireless communication practical.

The main idea

Transmit data on many orthogonal tones. Generate them with an IFFT. Add a cyclic prefix so multipath becomes a multiplication on each tone. Use an FFT and equalizer to recover the data.

Learning goals

- Connect Fourier series, continuous and discrete transforms, DFT and FFT.
- Calculate linear and circular convolution and derive the CP condition.
- Explain subcarrier orthogonality and the OFDM transmitter and receiver.
- Apply channel estimation, equalization and throughput calculations.
- Recognize synchronization, fading, noise and PAPR limitations.

Audience and study plan

For students and engineers with basic algebra and trigonometry. Complex numbers are reviewed next; calculus helps with the continuous-time formulas. Allow three 90-minute study sessions plus a simulation lab.

This guide teaches the essential foundation and introduces advanced topics. Numerical designs are illustrative, rather than requirements of a particular wireless standard.

1 / Signals, complex numbers and sampling

Complex baseband and I/Q

A complex sample is z = I + jQ, where $j^2 = -1$. Its magnitude is $\sqrt{(I^2+Q^2)}$ and its phase is atan2(Q,I). Euler's formula connects rotation and sinusoids:

$$e^{j\theta} = \cos\theta + j\sin\theta$$

The real RF waveform can be expressed as $s(t) = \mathrm{Re}\{x(t)e^{j2\pi fct}\}$. The complex envelope x(t) stores the in-phase and quadrature components. Complex baseband OFDM does not need a real-valued IFFT output.

Sampling

$$x[n] = x(nT_s), \quad T_s = 1/F_s$$

For a complex baseband spectrum confined to -B/2 through B/2, ideal sampling requires $F_s > B$. For a real lowpass signal with highest frequency B, use $F_s > 2B$. Practical filters require transition-band margin. Aliasing folds spectral replicas together and cannot be undone in general.

Energy, power and decibels

$$E = \Sigma_n|x[n]|^2, \quad P = (1/N)\Sigma_n|x[n]|^2$$
$$\text{Power ratio in dB} = 10\log_{10}(P_1/P_2)$$

These are discrete sample energy and average sample power. Physical energy also depends on sample duration and impedance. For equal impedance, amplitude ratios use $20\log_{10}(A_1/A_2)$.

Example: 1+j has magnitude $\sqrt{2}$, phase π/4 and squared magnitude 2. A 2× amplitude increase gives 4× power, or about 6.02 dB.

Notation

N: FFT size; n: time index; k: frequency bin; X[k]: subcarrier symbol; x[n]: time sample; G: prefix samples; h[ℓ]: channel taps; L: tap count; H[k]: channel DFT; * as a superscript: complex conjugate.

2 / Fourier theory: four related representations

Fourier series: periodic continuous-time signals

$$x(t) = \Sigma_{k=-\infty}^{\infty} c_k e^{j2\pi kt/T_0}$$

$$c_k = (1/T_0)\int_0^{T_0} x(t)e^{-j2\pi kt/T_0}dt$$

A signal of period T_0 is decomposed into harmonics separated by $1/T_0$, subject to the usual convergence conditions.

Continuous-time Fourier transform (CTFT)

$$X(f) = \int_{-\infty}^{\infty} x(t)e^{-j2\pi ft}dt$$

$$x(t) = \int_{-\infty}^{\infty} X(f)e^{j2\pi ft}df$$

The frequency variable is continuous and measured in hertz. Ideal tones require impulse representations. Time multiplication and convolution have corresponding dual operations in frequency.

Discrete-time Fourier transform (DTFT)

$$X(e^{j\omega}) = \Sigma_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$$

The DTFT has continuous frequency ω in radians/sample and repeats every 2π. The inverse integrates over any interval of length 2π with factor $1/(2\pi)$. Physical frequency is $f = \omega F_s/(2\pi)$.

Discrete Fourier transform (DFT)

For an N-sample block taken as zero outside the block, the DFT samples its DTFT at $\omega_k = 2\pi k/N$. The inverse DFT can also be viewed as a periodic extension of the block. That periodic interpretation leads to circular convolution.

Keep the distinctions clear: sampling time does not automatically make frequency discrete. Fourier series has discrete harmonics; the DTFT of a discrete-time sequence still has a continuous frequency argument.

3 / DFT mathematics and the FFT algorithm

$$X[k] = \Sigma_{n=0}^{N-1} x[n] e^{-j2\pi kn/N}$$

$$x[n] = (1/N) \Sigma_{k=0}^{N-1} X[k] e^{j2\pi kn/N}$$

We use an unnormalized forward DFT and a 1/N inverse. The forward transform projects onto tones; the inverse adds tones with weights X[k].

Worked four-point example

For X = [1, 1, -1, -1], the IDFT is x = [0, (1+j)/2, 0, (1-j)/2]. For n=1, the numerator is 1+j+1+j; divide by 4. Taking the DFT recovers X.

FFT: faster computation of exactly the same DFT

For even N, let E[k] and O[k] be N/2-point DFTs of the even and odd samples. With $W_N = e^{-j2\pi/N}$:

$$X[k] = E[k] + W_N^k O[k]$$

$$X[k+N/2] = E[k] - W_N^k O[k]$$

This is a butterfly. Recursion for $N=2^m$ yields m stages. Direct DFT complexity is $O(N^2)$; FFT complexity is $O(N \log_2 N)$. FFT algorithms also exist for other lengths.

Important properties

- Linearity: a x + b v transforms to a X + b V.
- Circular time shift by m gives phase $e^{-j2\pi km/N}$.
- Parseval: $\Sigma|x[n]|^2 = (1/N)\Sigma|X[k]|^2$.
- Bins above N/2 represent negative frequencies in wrapped ordering.
- Zero padding refines the plotted frequency grid; it does not add observation time or true resolving ability.

Implementation check: FFT(IFFT(X)) must equal X to numerical precision. Use fftshift mainly for plotting; keep transmitter and receiver bin ordering consistent.

4 / Linear convolution and multipath

A channel adds delayed copies

For a linear time-invariant finite impulse response channel with L taps:

$$y[n] = \Sigma_{\ell=0}^{L-1} h[\ell] x[n-\ell] + w[n]$$

This is linear convolution plus additive noise. An input of N samples produces N+L-1 convolution samples. The tail can overlap the next symbol, creating intersymbol interference (ISI).

Worked example

Let x = [1,2,3,4] and h = [1,0.5]. The channel adds half of the previous sample:

$$x*h = [1, 2.5, 4, 5.5, 2]$$

For instance, y[2] = 3 + 0.5×2 = 4. The last value 2 is a delayed contribution from the final sample.

Frequency-selective fading

$$H(e^{j\omega}) = \Sigma_{\ell=0}^{L-1} h[\ell] e^{-j\omega\ell}$$

Paths add constructively at some frequencies and destructively at others. The resulting amplitude and phase vary across the occupied band. OFDM makes this channel easier to equalize when the channel is sufficiently constant during one symbol.

Delay spread and coherence

Delay spread describes dispersion in arrival times. Coherence bandwidth describes the frequency range over which the channel is strongly correlated; it generally decreases as delay spread increases, with the numerical relationship depending on the definition. Doppler measures time variation; high Doppler reduces coherence time.

Assumptions: the basic derivation uses a finite L-tap channel, approximately time invariant during the symbol. Fractional delays and transmit/receive filtering may extend the effective impulse response. L is tap span, not simply the number of physical paths.

5 / Circular convolution and orthogonality

Circular convolution wraps sample indices

$$y_c[n] = \Sigma_{\ell=0}^{N-1} h[\ell]x[(n-\ell) \bmod N]$$

Zero-pad h to N samples when L≤N. For x=[1,2,3,4] and h=[1,0.5,0,0], circular convolution is [3,2.5,4,5.5]. Its first value is 1+0.5×4=3: the linear tail wraps to the start.

The DFT convolution theorem

Insert the circular sum into the DFT and substitute m=(n-ℓ) mod N. The periodic exponential allows the sums to separate:

$$Y[k] = (\Sigma_\ell h[\ell]e^{-j2\pi k\ell/N})(\Sigma_m x[m]e^{-j2\pi km/N})$$
$$Y[k] = H[k]X[k]$$

To calculate ordinary linear convolution with FFTs, instead zero-pad both sequences to a transform length at least N+L-1.

Orthogonal tones

$$\Sigma_{n=0}^{N-1} e^{j2\pi(k-m)n/N} = N \text{ for } k=m;\ 0 \text{ otherwise}$$

For distinct bins, the geometric-series ratio $r=e^{j2\pi(k-m)/N}$ satisfies $r^N=1$ and $r\neq1$, so the sum $(1-r^N)/(1-r)$ is zero. Continuously, the normalized integral of $e^{j2\pi(k-m)t/T_u}$ over $0\le t\le T_u$ gives the same orthogonality.

$$T_u = N/F_s, \quad \Delta f = F_s/N = 1/T_u$$

Rectangular useful symbols have sinc-shaped spectra. Adjacent centers coincide with zeros of other tones, allowing spectral overlap without interference under ideal alignment. Frequency offset and time variation can destroy this cancellation.

6 / Cyclic prefix: the key derivation

Copy the final G samples to the front

[x[N-G] ... x[N-1] | x[0] ... x[N-1]]

cyclic prefix | useful N-sample symbol

Define $x_{CP}[n]=x[n \bmod N]$ for $-G \le n \le N-1$. The receiver discards the prefix and keeps the N useful samples.

Why it works

Assume correct timing and $G \ge L-1$. For each retained $n=0,\ldots,N-1$, every sample $x[n-\ell]$ required by the channel belongs to the same symbol's cyclic extension. Thus:

$$y[n] = \Sigma_{\ell=0}^{L-1} h[\ell]x[(n-\ell) \bmod N] + w[n]$$
$$\text{FFT} \Rightarrow Y[k] = H[k]X[k] + W[k]$$

At n=0, the oldest needed sample is $x_{CP}[-(L-1)]$. This directly gives the minimum prefix length in the ideal model.

Numerical check

For x=[1,2,3,4], h=[1,0.5] and G=1, transmit [4 | 1,2,3,4]. Filter it and retain positions corresponding to [1,2,3,4]. They are [3,2.5,4,5.5], precisely the circular convolution from page 6.

Overhead and limitations

$$T_{CP}=G/F_s, \quad T_{sym}=(N+G)/F_s$$
$$\eta_{CP}=N/(N+G)$$

The prefix adds energy and time but no new payload. At fixed average transmit power, its payload energy efficiency penalty is $-10\log_{10}\eta_{CP}$ dB. Timing uncertainty and filter transients need design margin.

The physical channel still performs linear convolution. The copied prefix and selected receive window produce the circular relationship. The prefix does not remove noise, CFO or within-symbol channel variation. Zero padding alone does not give the same simple FFT model.

7 / Transmitter and receiver processing

Bits → coding / interleaving → QAM → data/pilot bins
→ IFFT → CP → DAC / filtering → RF

Transmitter

QPSK maps two bits to $(\pm 1 \pm j)/\sqrt{2}$ for unit symbol energy. M-QAM carries $\log_2 M$ bits per mapped symbol. Coding adds redundancy; interleaving spreads coded bits across tones and time to reduce concentrated losses from fading.

Allocate payload to data bins and known reference symbols to pilot bins. Edge nulls provide guard bands; DC is often nulled to mitigate DC-related impairments. The IFFT synthesizes the waveform, and the prefix is added before serialization.

With K independent zero-mean unit-energy active tones and a 1/N IFFT, $E\{|x[n]|^2\}=K/N^2$. Set waveform power with a known gain and account for that gain in reception.

RF / ADC → packet and frequency synchronization
→ remove CP → FFT → estimate H → equalize → demap / decode

Receiver

The receiver detects the frame, corrects frequency error, selects the FFT window and removes the prefix. After the FFT, channel estimation provides $\hat{H}[k]$ for equalization.

$$\text{ZF: } \hat{X}[k]=Y[k]/H[k]$$
$$\text{MMSE: } \hat{X}[k]=H^*[k]Y[k]/(|H[k]|^2+\sigma^2{}_W/E_s)$$

ZF reverses channel amplitude and phase, but amplifies noise at small |H|. Scalar linear MMSE assumes known H, independent zero-mean symbols of variance E_s and uncorrelated noise variance $\sigma^2{}_W$. It balances attenuation and noise enhancement.

If time noise variance is $\sigma^2{}_w$, the unnormalized FFT yields $\sigma^2{}_W=N\sigma^2{}_w$. Generate complex AWGN with real and imaginary variances $\sigma^2{}_w/2$ each. Consistent normalization is essential.

8 / Synchronization, pilots and practical limits

Timing and frequency synchronization

A preamble or repeated sequence supports packet detection, timing and coarse frequency estimation. The FFT window must stay in the valid interval allowed by the cyclic extension and channel delays. A shift inside that interval gives a phase slope across tones; a shift outside introduces ISI and ICI.

$$r[n]=y[n]e^{j2\pi\varepsilon n/N},\quad \varepsilon=\delta f/\Delta f$$

Noninteger normalized carrier frequency offset ε causes intercarrier interference (ICI). An ideal integer offset shifts bin indices. Estimate and remove CFO before the FFT; pilots can track residual common phase error.

Pilot channel estimation

$$\hat{H}[k_p] = Y[k_p]/X[k_p]$$

This least-squares estimate uses a known pilot $X[k_p]$. Interpolate across time and frequency. Pilot spacing must resolve channel variation due to delay spread and Doppler. Averaging reduces noise only while the channel remains sufficiently stable.

PAPR and spectral containment

$$\mathrm{PAPR} = \max |x(t)|^2 \,/\, E\{|x(t)|^2\}$$

Subcarriers can align to form large peaks; K equal-amplitude tones can reach a theoretical ratio K over a useful interval. Amplifier backoff avoids distortion but reduces efficiency. Clipping causes distortion and out-of-band radiation; tone reservation sacrifices some carriers. Oversampling is needed to estimate intersample peaks.

Other limitations and extensions

Phase noise, sample clock error, I/Q imbalance and rapid fading degrade orthogonality. Windowing/filtering improve spectral containment but add transients or overhead. OFDMA assigns time-frequency resources to users. In MIMO, each tone has a matrix model $\mathbf{Y}[k]=\mathbf{H}[k]\mathbf{X}[k]+\mathbf{W}[k]$; spatial equalization is still required. DFT-spread OFDM adds DFT precoding and can reduce envelope variation.

9 / Worked design and rate calculations

Use N=64, F_s=1 MHz, G=8, 48 data tones, 4 pilot tones and 12 nulls. Choose QPSK with code rate R_c=1/2.

Quantity	Calculation	Result
Sample period	$1/F_s$	1 µs
Useful duration	N/F_s	64 µs
Tone spacing	F_s/N	15.625 kHz
Prefix duration	G/F_s	8 µs
Total duration	$(N+G)/F_s$	72 µs
CP efficiency	64/72	88.89%
Coded bits / symbol	48×2	96
Information bits / symbol	96×1/2	48
Payload rate estimate	48/(72 µs)	0.667 Mbit/s

$$R \approx N_{data} \log_2(M) R_c / T_{sym}$$

Preambles, control, scheduling, retransmissions and higher-layer headers reduce delivered throughput. This formula assumes every counted tone uses the stated modulation and code rate.

Check delay coverage

A latest modeled tap at 6 µs relative delay means tap indices 0 through 6 at 1 µs/sample: L=7. G=8 exceeds L-1=6, leaving two samples of ideal margin. Include filtering and timing uncertainty in the real margin.

Design tradeoff

Doubling N at fixed F_s and G halves spacing and doubles useful duration. CP efficiency improves to 128/136=94.12%, while symbol latency and sensitivity to normalized frequency error increase. Larger spacing shortens symbols and can improve tolerance to time variation.

52×Δf=812.5 kHz is a nominal active-bin bandwidth measure. Actual occupied bandwidth depends on tone allocation, sidelobes, filtering and the measurement definition.

10 / Simulation lab: verify the entire chain

Run with Python and NumPy. The model uses perfect timing and known channel taps. It deliberately omits coding and pilot estimation.

```
import numpy as np
rng = np.random.default_rng(7)
N, G = 64, 8
active = np.r_[np.arange(1,27), np.arange(38,64)]
bits = rng.integers(0,2,(len(active),2))
s = ((1-2*bits[:,0])+1j*(1-2*bits[:,1]))/np.sqrt(2)
X = np.zeros(N,complex)
X[active] = s
x = np.fft.ifft(X)
tx = np.r_[x[-G:],x]
h = np.array([1,0.3+0.2j,0,0.1j])
rx = np.convolve(tx,h)
H = np.fft.fft(h,N)
Y0 = np.fft.fft(rx[G:G+N])
assert np.max(np.abs(Y0-H*X)) < 1e-12

# Mean receive SNR over active FFT bins = 20 dB.
Ps = np.mean(np.abs(H[active]*s)**2)
nv = Ps/(N*10**(20/10))
w = np.sqrt(nv/2)*(rng.standard_normal(len(rx))
                   +1j*rng.standard_normal(len(rx)))
Y = np.fft.fft((rx+w)[G:G+N])
Z = Y[active]/H[active]
detected = np.column_stack((Z.real<0,Z.imag<0))
print('BER:',np.mean(detected != bits))
print('RMS EVM:',np.sqrt(np.mean(np.abs(Z-s)**2)))
```

Experiments and expected observations

1. Remove noise: the assertion verifies Y=HX and bit recovery should be exact.
2. Repeat many symbols and sweep SNR: BER should fall as SNR rises. One block has only 104 bits, insufficient for a reliable low BER estimate.
3. For insufficient CP, concatenate multiple symbols before filtering; decode each at its own offset. This captures previous-symbol interference.
4. Apply $e^{j2\pi 0.05n/N}$ to received samples: observe ICI.
5. Plot |H| and compare ZF with MMSE near weak tones.

11 / Exercises, answers and further study

Practice questions

1. N=128 and F_s=2 MHz: find tone spacing and useful duration.
2. L=10 and G=8: does the prefix cover the ideal channel?
3. Find the four-point DFT of [1,0,0,0].
4. Circularly convolve [1,2,3,4] with [1,-1,0,0].
5. N=256, G=32, F_s=4 MHz: find CP efficiency and total duration.
6. 100 data tones, 16-QAM, rate 3/4 coding, 100 µs symbol duration: estimate information rate.

Worked answers

1. Δf=2 MHz/128=**15.625 kHz**; T_u=**64 µs**.
2. **No.** Need G≥L-1=9 samples under ideal timing.
3. **[1,1,1,1]**; only the n=0 sample contributes.
4. y[n]=x[n]-x[(n-1) mod 4]=**[-3,1,1,1]**.
5. 256/288=**88.89%**; 288/(4 MHz)=**72 µs**.
6. 100×4×0.75/(100 µs)=**3 Mbit/s** before extra overhead.

Quick glossary

AWGN: additive white Gaussian noise. **BER:** bit error rate. **EVM:** error vector magnitude. **ICI / ISI:** intercarrier / intersymbol interference. **QAM:** quadrature amplitude modulation. **ZF:** zero forcing. **MMSE:** minimum mean-square error.

Further study

Established textbooks for deeper reading (the guide is self-contained):

- Oppenheim and Schafer, *Discrete-Time Signal Processing*: transforms, sampling and convolution.
- Proakis and Salehi, *Digital Communications*: modulation, detection, coding and equalization.
- Tse and Viswanath, *Fundamentals of Wireless Communication*: fading, OFDM and MIMO.
- van Nee and Prasad, *OFDM for Wireless Multimedia Communications*: multicarrier design.

Final checkpoint: derive why a sufficient CP makes the retained samples a circular convolution, and then show why the FFT yields Y[k]=H[k]X[k]+W[k].